# SHELF & SOUL

Technical Project Report | Brief overview | Prepared 19 October 2026

## Project summary

Shelf & Soul is a library and bookshop management application built with Next.js and TypeScript. The browser-based development prototype is available at **http://localhost:3000**. The production design uses server APIs and MySQL; production readiness depends on completing environment and operations setup described below.

| Area | Implementation |
| --- | --- |
| Application | Next.js 15, React 19, TypeScript; Next.js API routes for server-side operations. |
| Prototype | Browser Local Storage for demo data and local-only authentication. Intended for demonstration, not production access control. |
| Production data | MySQL through Prisma 6. Staff/user records are relational; most library records currently reside in a JSON LibrarySnapshot. |
| Authentication | bcrypt password hashing, signed HTTP-only sessions, and role-checked server API actions. |

## Implemented capabilities

- Catalog and member management; staff roles and permissions; borrowing, returns, reservations, and renewals.
- Inventory adjustments, damaged/lost stock, location transfers, suppliers, purchase orders, and invoice capture.
- Sales and pickup orders, fine recording, audit history, and filtered CSV/print reports.
- Manual manager-controlled library snapshot export/restore; encrypted MySQL backup command with retention support.

## Payment handling (included at the user's request)

No online payment processor is integrated. Customer book orders use **cash on pickup**: orders remain unpaid until staff confirms collection and cash receipt. In-library sales are cash sales, and staff can record cash fine payments. This describes the implemented flow, not a payment gateway.

## Password recovery

Production reset uses one-time, hashed tokens that expire after 30 minutes; the response is designed not to reveal whether an account exists. Reset email requires a configured provider endpoint and deployment secrets. The prototype shows a local reset code and does not send email.

## Validation and local check

- Passed: Prisma client generation and schema validation, TypeScript typecheck, production build, and backup script syntax check.
- Browser smoke check: exercised the prototype password-reset flow and subsequent sign-in; localhost returned HTTP 200.
- Not verified against a live production database or email provider; no end-to-end scheduled backup/decryption/restore test was run.

## Production readiness and remaining work

- Configure MySQL and AUTH_SECRET, apply migrations, and test production APIs against the actual database.
- Configure the email provider and trusted HTTPS APP_BASE_URL; test delivery and password recovery.
- Set up Windows Task Scheduler, protected backup key/destination, off-site copy, and a tested recovery procedure. The backup command does not implement decryption/restore.
- Email verification, migration from JSON snapshot to normalized relational tables, and real branch contact/hours remain incomplete or deployment-dependent.

**Conclusion:** the local prototype runs and the recorded code/build checks passed. This is not a claim that every SRS requirement is complete or that a production deployment has been tested.